extranuclear atomic structure we stated that the local density of the matter waves constitutes a measure of the probability of encountering alpha particles at the point under consideration. But this density outside of the nucleus, and hence the probability of finding alpha particles outside of it, increases rapidly when we deal with a particle which is rich in energy, and more slowly when our particle is less rich in energy. The more energy a particle has, the greater is the probability that it will no longer be found inside, but outside of the nucleus—in other words, that it is hurled out within a very short time. This is where we find the explanation of the fact that in the case of alpha particles rich in energy the decay probability is very much greater than in the case of those less rich in energy. The exact mathematical expression of this idea leads to a very satisfactory agreement with the Geiger-Nuttall law.

If we wish to give a quite summary description, in terms of the particle aspect, of the effect just discussed, we may state that contrary to all expectations based on the law of the conservation of energy, after a certain period of time (the length of which is a matter of chance) the particle is able to break through the potential barrier as though through a tunnel. It is customary therefore to speak of a *tunnel effect.*

We have already mentioned that although all the elements above zinc in the periodic system might be expected to be prone to alpha decay, this decay can, nevertheless, be observed in the heaviest elements only. It would be logical to assume that no sharp dividing line existed between the well-known alpha emitters and those elements which would seem capable, from the point of view of energetics, of emitting alpha radiation, but are never observed to do so. It may well be possible that a major number of the elements above zinc in the periodic table do actually emit alpha particles, but both the energy and range of these particles are very small. Both these factors would account amply for our practical inability to observe them. However, such radioactivity would be imperceptible mainly because the low magnitude of energy corresponds to an extremely small decay probability. For even though such an element might emit an alpha particle at infrequent intervals, it is nevertheless stable for all practical purposes, even when measured by cosmical time standards.

II. THE BETA EMITTERS

Let us now turn our attention to the second type of spontaneous nuclear transmutation, which takes place with beta radiation—the emission of electrons or positrons, accompanied by a neutrino. Such transmutations take place when they are compatible with the conservation laws, and thus in particular when energy is liberated from the nucleus in the process. Here, too,